

Rajvardhan Patil

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EDUCATION

University of Colorado Boulder

Boulder, CO, USA

Master of Science in Computer Science

August 2025 – May 2027

Coursework: Neural Networks and Deep Learning, Big Data Architecture, Database Systems, Data Mining, Quantum Computing

Vellore Institute of Technology (VIT)

Vellore, India

Bachelor of Technology in Computer Science and Engineering; GPA: 3.4/4.0

July 2021 – May 2025

EXPERIENCE

University of Colorado Boulder – Institute of Cognitive Science

Boulder, CO, USA

Software Engineer (Student Assistant)

May 2026 – Present

- Contribute to a **Flask/MongoDB** Unified Verb Index ([link](#)), serving VerbNet, PropBank, FrameNet, and OntoNotes to NLP researchers; run **2 production Linux servers** at **99.99% uptime over 18,772 health checks** with zero-downtime gunicorn reloads under **systemd**.
- Rebuilt ingest as a Python **ETL pipeline**: **NLTK** parses FrameNet, **spaCy** dependency parses generate syntactic frame diagrams, and cross-links across **4 corpora** (3K+ entries) are computed once into **denormalized** collections, replacing request-time joins with single-document reads.
- Cut p95 search and page latency from 897ms to 287ms across **20+ views** by replacing full collection scans with **MongoDB indexes** and moving repeated **Jinja2** lookups into precomputed fields; patched OpenSSL CVEs and upgraded to **Flask 3 / Python 3.12**, gated by **pytest** in **GitHub Actions** CI.

University of Colorado Boulder – Civil & Architectural Engineering

Boulder, CO, USA

Software Engineer (Student Assistant)

March 2026 – Present

- SABER ([link](#))**: Led the rewrite of a PyQt5 desktop energy-audit tool into a web app with 2 co-developers: a **FastAPI** backend on **Azure** runs the change-point regression engine (**pandas/NumPy**) behind **async** endpoints so long fits don't block other requests, a **RAG** pipeline handles standards lookups, and a **React/TypeScript** frontend plots energy use against weather in **Plotly**.
- Panel Sizer ([link](#))**: NEC Article 220 residential load calculator in **FastAPI**, **Dockerized** and deployed on **Azure Container Apps** with infrastructure as code in **Terraform**, plus a **RAG** assistant for NEC code questions.
- TUNBEEC Compliance Tool**: Wrapped the existing Tunisian energy-code engine, unmodified, in a thin **FastAPI** layer, built a **Next.js** frontend at parity with the PySide6 desktop app on **Azure App Service**, and shipped an offline Windows installer (PyInstaller).

Walstar Technologies

Kolhapur, India

Software Engineering Intern

August 2023 – October 2023

- Built a dairy cooperative app (**Flutter, Laravel, MySQL**) replacing paper records; adopted by **5+ cooperatives**, saving each **7 hours/week** (about 40% of manual work).
- Designed the **PHP/MySQL REST APIs** and Figma prototypes, and shipped the Android frontend with **95% UI test coverage**.

VOLUNTEER & LEADERSHIP

Computer Association of Kolhapur (CAK)

Kolhapur, India (Remote)

Software Developer (Volunteer)

January 2026 – Present

- Built the membership portal (**Next.js, FastAPI, Supabase/PostgreSQL**) that replaced manual signups for 30+ members: multi-step onboarding and **Razorpay** payments, where payment confirmation triggers membership activation and an auto-generated **PDF receipt**.
- Built the admin dashboard, secured by Firebase-issued **JWTs** verified server-side in **FastAPI**, with live membership and revenue stats and a bulk **Excel import** that validates and deduplicates member records before insert.

VinnovateIT

Vellore, India

Application Developer Lead

November 2022 – December 2024

- Led development of **MessIt (20,000+ users)** and **Bunkbuddies (React, Node.js, MongoDB)**; built a shared component library and added **end-to-end tests** to the release process, cutting recurring bug reports from 8 to 0 per month.

PROJECTS

- Gradmits ([link](#))**: Shortlists US Master's programs from **250K+ real admission decisions**, with LLM recommendations, visa interview insights, faculty discovery, and application tracking. Tech: **Next.js, FastAPI, MongoDB, AWS**.
- GradBro ([link](#))**: Writing copilot for graduate Statements of Purpose used by **1,500+ applicants**. A **FastAPI** backend calls **Claude** and **OpenAI** to suggest edits rather than rewrite, so the essay stays in the applicant's voice. Tech: **Next.js, FastAPI, AWS CloudFront**.
- Manter ([link](#))**: Dating PWA that matches on character, not photos. A 14-scenario quiz is scored deterministically across **23 traits** and weighted by each user's priorities; realtime chat runs a **Claude Haiku** red-flag scan through an **idempotent** endpoint that re-fetches messages server-side instead of trusting the client. Tech: **Next.js, TypeScript, FastAPI, Supabase** (Postgres, Realtime).
- Pumped Up Kicks ([link](#))**: Team of 4. Q&A over recorded lectures: **Whisper** transcribes with timestamps, chunks are embedded in **pgvector**, and top-k semantic retrieval sends only matching excerpts to Claude; answers stream over **SSE** and link to the exact moment in the video. Tech: **Next.js, FastAPI, PostgreSQL/pgvector, Alembic, Clerk**.

RESEARCH

- Smart Refrigerator Model for Food Safety and Health Promotion ([link](#))**: Built a **YOLOv10**-based system for real-time food identification and spoilage detection with **97.5% accuracy**, plus an Android app for replenishment alerts. Published in *AIP Conf. Proc.* 3388.

TECHNICAL SKILLS

- Languages/Frontend**: Python, TypeScript, JavaScript, SQL, C++, Java, Bash, PHP; React, Next.js, Tailwind CSS, Plotly, Flutter.
- AI/ML**: LLM APIs (Claude, OpenAI), RAG, Vector Embeddings, Semantic and Hybrid Search (BM25), Ranking, pgvector, Whisper, YOLO, spaCy, NLTK, pandas, NumPy, LightGBM, Model Calibration.
- Cloud/Infra**: Azure (App Service, Container Apps), AWS (S3, CloudFront, Lambda), Google Cloud (Cloud Run, Pub/Sub, GKE), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Git, Linux, systemd.
- Backend/Data**: FastAPI, Flask, Node.js, Laravel, REST APIs, SQLAlchemy, Alembic, PostgreSQL, MongoDB, MySQL, Supabase, Redis, Spark, MinIO.
- Concepts**: Event-driven systems (queues, idempotency, retries with backoff, bounded concurrency), async I/O, indexing and denormalization, caching, ETL, IaC, hybrid retrieval and chunking, ML evaluation (leakage, calibration, precision/recall), DSA.